WORK SHEET - 03 **ONIC EQUILIBRIUM**

Single correct

Q.1 Three solution of strong electrolytes 25 ml of 0.1 M HX, 25 ml of 0.1 M H_2Y and 50 ml of 0.1 $NZ(OH)_2$ are mixed. What is the pOH of the solution ? [Given log 5 = 0.7]
(A) 1.6 (B) 7 (C) 11.6 (D) 12.6

Q.2 0.1 M NH_3 (aq) solution is titrated against 0.1 M HNO_3 (aq) solution. What would be the difference in pOH between $\frac{1}{4}$ and $\frac{3}{4}$ stages of neutralization of base?
(A) 2 log 3/4 (B) 2 log 1/4 (C) log 1/3 (D) 2 log 3

Q.3 What is the pH of 0.1M $NaHCO_3$? $K_1 = 5.0 \times 10^{-7}$, $K_2 = 7.0 \times 10^{-11}$ for carbonic acids.
(A) pH = 7 (B) pH much less than 7
(C) pH slightly less than 7 (D) pH greater than 7

Q.4 The degree of hydrolysis of 0.1 M solution of conjugate base of HA is 0.01. Find the $\mathbf{H^+}$ concentration in 0.4 M solution of $\mathbf{A^-}$.
(A) 5×10^{-11} M (B) 5×10^{-12} M (C) 2×10^{-3} M (D) 2×10^{-4} M

Q.5 What is the aq. ammonia concentration of a solution prepared by dissolving 0.15 mole of $NH_4^+CH_3COO^-$ in 1L H_2O.[$K_{a(CH_3COOH)} = 1.8\times10^{-5}$; $K_{b(NH_4OH)} = 1.8\times10^{-5}$]
(A) 8.3×10^{-4} (B) 0.15 (C) 6.4×10^{-4} (D) 3.8×10^{-4}

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Q.6 An aqueous solution contains 0.01 M RNH_2 ($K_b = 2 \times10^{-6}$) & 10^{-4} M NaOH.
The concentration of OH^- is nearly :
(A) 2.414×10^{-4} (B) 10^{-4} M (C) 1.414×10^{-4} (D) 2×10^{-4}

Q.7 At 90°C, pure water has $[H^+]=10^{-6}$M, if 100 ml of 0.2 M HNO_3 is added to 20 ml of 1 M NaOH at 90°C then pH of the resulting solution will be
(A) 5 (B) 6 (C) 7 (D) None of these

Q.8 Calcium lactate is a salt of weak organic acid and strong base represented as $Ca(LaC)_2$. A saturated solution of $Ca(LaC)_2$ contains 0.6 mole in 2 litre solution. pOH of solution is 5.60. If 90% dissociation of the salt takes place then what is pK_a of lactic acid
(A) 2.8 – log(0.54) (B) 2.8 + log(0.54) (C) 2.8 + log (0.27) (D) none

Q.9 What is the degree of dissociation of weak acid HA (C = 0.1 M) in presence of strong acid HB (C = 0.1M). Given : K_a (weak acid) = 10^{-6}
(A) 10^{-5} (B) 10^{-6} (C) 10^{-4} (D) 10^{-3}

Q.10 2.5 mL of $\frac{2}{5}$ M weak monoacidic base ($K_b = 1 \times 10^{-12}$ at 25°C) is titrated with $\frac{2}{15}$ M HCl in water at 25°C. The concentration of H^+ at equivalence point is
($K_w = 1 \times 10^{-14}$ at 25°C)
(A) 3.7×10^{-13} M (B) 3.2×10^{-7} M (C) 3.2×10^{-2} M (D) 2.7×10^{-2} M

Q.11 An aqueous solution of CH_3COOH has a pH = 3 and acid dissociation constant of CH_3COOH is 10^{-5}. What will be the concentration of acid taken initially?
(A) 0.1M (B) 0.11 M (C) 0.09 M (D) 0.101 M

Q.12 pH of a saturated solution of silver salt of monobasic acid HA is found to be 9.
Find the K_{sp} of sparingly soluble salt Ag A(s). **Given :** K_a (HA) = 10^{-10}
(A) 1.1×10^{-11} (B) 1.1×10^{-10} (C) 10^{-12} (D) None of these

Q.13 Which of the following expression regarding % ionisation of a monobasic weak acid in aqueous solution at appreciable concentration is not correct?

(A) $100\sqrt{\frac{K_a}{C}}$ (B) $\left\{\frac{K_a}{K_a+[H^+]}\right\}\times 100$ (C) $\left\{\frac{[H^+]}{K_a+[H^+]}\right\}\times 100$ (D) $\frac{100}{1+10}(pK_a - pH)$

Q.14 The pH of a saturated solution of $X(OH)_3$ will be if its solubility product = 2.7×10^{-43}.
[**Given :** log 3 = 0.5 , log 2 = 0.3]
(A) 10.5 (B) 7 (C) 3.5 (D) 3.8

Q.15 Let solubility of a sparingly soluble salt of strong base and weak acid in pure water is equal to $\mathbf{s_1}$, in a buffer solution with pH < 7 is equal to $\mathbf{s_2}$ and in another buffer solution with pH > 7 is equal to $\mathbf{s_3}$. If hydrolysis is considered in all the solutions then the **correct** relation between $\mathbf{s_1}$, $\mathbf{s_2}$ and $\mathbf{s_3}$ is :
(A) $s_1 = s_2 = s_3$ (B) $s_1 > s_2 > s_3$ (C) $s_1 < s_2 < s_3$ (D) $s_2 > s_1 > s_3$

Q.16 The salt $Zn(OH)_2$ is involved in the following two equilibria,
$Zn(OH)_2(s) \rightleftharpoons Zn^{2+}(aq) + 2OH^-(aq)$; $K_{sp} = 10^{-17}$
$Zn(OH)_2(s) + 2OH^-(aq) \rightleftharpoons [Zn(OH)_4]^{2-}(aq.)$; $K_c = 0.1$
The pH of solution at which solubility is minimum is
(A) 10 (B) 9 (C) 8 (D) 11

Q.17 From separate solutions of four sodium salts NaW, NaX, NaY and NaZ had pH 7.0, 9.0, 10.0 and 11.0 respectively. When each solution was 0.1 M, the strongest acid is:
(A) HW (B) HX (C) HY (D) HZ

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Q.18 How many moles NH_3 must be added to 2.0 litre of 0.80 M $AgNO_3$ in order to reduce the Ag^+ concentration to 5×10^{-8} M. K_f of $[Ag(NH_3)_2^+] = 10^8$
(A) 0.4 (B) 2 (C) 3.52 (D) 4

Q.19 A_3B_2 is a sparingly soluble salt of molar mass M ($g\,mol^{-1}$) and solubility x g lit^{-1}. The ratio of the molar concentration of B^{3-} to the solubility product of the salt is

(A) $108\frac{x^5}{M^5}$ (B) $\frac{1}{108}\frac{M^4}{x^4}$ (C) $\frac{1}{54}\frac{M^4}{x^4}$ (D) None

Assertion and Reason

Q.20 **Statement-1**: pH of acidic buffer solution always increases on increasing dilution.
Statement-2: pH of any acidic solution always increases on increasing dilution.
(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.
(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.
(C) Statement-1 is true, statement-2 is false. (D) Statement-1 is false, statement-2 is true.

Comprehension :

Paragraph for question nos. 21 to 23

Solubility is defined as the maximum number of moles of solute which can be dissolved in 1 litre of solvent. It is affected by presence of common ion, hydrolysis of ions. In a container equal volume of 0.02 M $AgNO_3$ and 0.02 HCN solution is mixed.
K_{sp} of AgCN = $4\times10^{-16}\ M^2$,
K_a (HCN) = 4×10^{-10} M

Q.21 Concentration of Ag^+ ions in the final solution would be
(A) 10^{-6} (B) 2×10^{-5} (C) 2×10^{-8} (D) 10^{-4}

Q.22 pH of final solution would be
(A) 2 (B) 3 (C) 4 (D) Can not find

Q.23 Concentration of CN^- in the final solution would be
(A) 4×10^{-10} (B) 2×10^{-11} (C) 4×10^{-12} (D) 2×10^{-8}

Paragraph for question nos. 24 to 25

Selective precipitation of ions from a mixture in the form of salts can be done by adding common ion gradually. Let us consider selective precipitation of Cl^- & CrO_4^{2-} ions in the form of AgCl & Ag_2CrO_4 from a mixture having 0.01 M Cl^- and 0.02 M CrO_4^{2-}. For this, salt having Ag^+ [like $AgNO_3$(s)] is added gradually.

Given : K_{sp} AgCl = 10^{-10} and K_{sp} $Ag_2CrO_4 = 4 \times 10^{-14}$.

Q.24 Minimum concentration of Ag^+ ion at which precipitation of at least one ion starts.
(A) 10^{-8} (B) 1.41×10^{-6} (C) 10^{-10} (D) 2×10^{-8}

Q.25 The percentage of one of the ion precipitated when another ion starts precipitation is

(Given : $\frac{1}{\sqrt{2}} = 0.7$)

(A) 98.7 % (B) 99.3 % (C) 97.6 % (D) 92.7 %

More than one correct :

Q.26 At 25°C , the dissociation constant of a weak monoprotic acid, HA **(K_a)** is **numerically equal** to the dissociation constant of its conjugate base, A^- **(K_b)**. Which of the following statement(s) is/are **correct**?
(A) The dissociation constant of the acid, HA, is 10^{-7}.
(B) The pH of 0.1 M aqueous solution of the acid, HA, is 4.0.
(C) The pH of 0.1 M aqueous solution of the conjugate base (A^-) is 10.0.
(D) The pH of an aqueous solution containing 0.1 M - HA and 0.01 M - HCl is 2.0.

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Match the column

Q.27 Match column I having components with column II having pH or pOH of solution.

	Column I	Column II	
(A)	Saturated solution of $A(OH)_2$ having $k_{sp} = 5 \times 10^{-31}$	(P)	7
(B)	Mixing equal volumes of 0.1 M CH_3COONa & 0.1 M XNO_3	(Q)	10
(C)	Solution obtained at $\frac{10}{11}$ of the equivalence point when 0.1 M XOH is titrated with HCl 0.1 M	(R)	5
		(S)	4

[Given data : K_h $X^+ = 10^{-5}$, $K_{a,CH_3COOH} = 10^{-5}$]

Subjective :

Q.28 A buffer of pH 9.26 is made by dissolving x moles of ammonium sulphate and 0.1 mole of ammonia into 100 mL solution. If pK_b of ammonia is 4.74, calculate value of x.

Q.29 50ml of a weak monoprotic acid was titrated with 0.1M solution of NaOH. If pH of the solution after adding 50 ml and 75 ml of base is 4.699 and 5 then calculate a four digit number **'abcd'** where :
ab = pK_a of the weak acid; **cd** = pH of the solution when 7.5 millimoles of HCl are added in the solution at the equivalence point without changing volume.[**Given :** log 2 = 0.301]

Q.30 A solution contains 0.01M Mg^{+2} and 0.1M Sr^{+2} and H_2CO_3 maintained at 0.05M. At what minimum & maximum pH will $SrCO_3$ will precipitate without any precipitation of $MgCO_3$.
[**Given:** $K_{sp}MgCO_3 = 2.5 \times 10^{-7}$, $K_{sp}SrCO_3 = 9 \times 10^{-9}$, $K_{a(overall)}$ $H_2CO_3 = 5\times10^{-16}$, log5 = 0.7, log 3 = 0.5] ***Express your asnwer as abcd where* ab** = Ten times minimum pH for the above condition; **cd** = Ten times maximum pH for the above condition.

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<u>*LIST OF IMPORTANT FORMULAS*</u>

* **Basic Relationships**

(a) $K_w = [H^+][OH^-] = 10^{-4}$ (at 25°C) , K_a (for water) $= \dfrac{K_w \times 18}{1000}$

(b) $\ln\dfrac{K_{w_2}}{K_{w_1}} = \dfrac{\Delta H}{R}\left[\dfrac{1}{T_1} - \dfrac{1}{T_2}\right]$; $\Delta H = \Delta H$ neutralise of SA and SB

(c) pH + pOH = pK_w = 14 at 25°C

(d) For weak conjugate acid / base pair

$pK_a + pK_b = pK_w$

(e) Ostwald dil law for an electrolyte AB, $\dfrac{C\alpha^2}{1-\alpha} = K$

<u>pH CALCULATUION</u>

1. pH of a solution containing strong monoprotic acid.

(i) at moderate conc. ($C \geq 10^{-6}$ M)

$pH = -\log C$

(ii) at low conc. ($C < 10^{-6}$ M)

$K_\omega = (C + x)(x)$

$[H^+] = C + x$ $x \rightarrow$ contribution of water

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2. pH of a solution containing weak monoprotic acid.

Case-I : *If $C \geq 10^{-6}$ M*

$K_a = \dfrac{x^2}{C - x}$ If $C >> x$ $pH = \dfrac{1}{2}(pK_b - \log C)$

Case-II : If $C < 10^{-6}$ M

Weak acid can be treated as a strong acid but contribution due to water cannot be neglected.

$[H^+] = c + x$

$K_\omega = (c + x)\,x$

3. pH of a solution containing more than one acid.

Case I : Mixture of SA + SA

$[HA] = c_1$

$[HB] = c_2$

$[H^+] = c_1 + c_2$

Depending upon the final [H^+] condition, Contribution from water may be neglected or taken.

Case II : Mixture of SA + WA

$[HCl] = c_1 \qquad [HA] = c_2$

$$HA \rightleftharpoons H^+ + A^-$$
$$c_2 - x \qquad c_1 + x \qquad x$$

$$K_a = \frac{(c_1 + x)x}{(c_2 - x)}$$

Case-III : Mixture of WA + WA

$$HA \rightleftharpoons H^+ + A^- \qquad K_1 = \frac{(x+y)x}{c_1 - x}$$
$$c_1 - x \qquad x + y \qquad x$$

$$HB \rightleftharpoons H^+ + B^- \qquad K_2 = \frac{(x+y)y}{c_2 - y}$$
$$c_2 - y \qquad x + y \qquad y$$

* For polyprotic acid $K_1 >> K_2 >> K_3$

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All the formulaes derived above are applicable for bases also. Only [H^+] will be replaced with [OH^-]

SALT HYDROLYSIS

Case-I : ***Salts of strong acids and strong bases*** do not undergo hydrolysis.

Neutral $\Rightarrow pH = 7$

Case-II : ***Salts of strong base and weak acid*** give a basic solution (pH>7) when dissolved in water.

e.g. : CH_3COONa, NaCN

$$K_b = \frac{K_\omega}{K_a} = K_h$$

$$K_h = K_h = \frac{K_\omega}{K_a} = \frac{(ch)^2}{c(1-h)}$$

if $\frac{K_h}{c} \le 10^{-3}$ then $h << 1$

$$pH = \frac{1}{2}(pK_\omega + pK_a + \log C)$$

In case of substantial hydrolysis the above expression will not be applicable

Case-III : ***Salts of a strong acids and weak bases*** give an acidic solution.

$$K_a = \frac{K_\omega}{K_b} = K_2$$

$$\frac{K_\omega}{K_b} = \frac{(ch)^2}{c(1-h)} = K_h$$

if $\quad \frac{K_h}{c} \le 10^{-3} \qquad h << 1$

$$= \frac{1}{2}\left(-\log K_\omega + \log K_b - \log C\right) = \frac{1}{2}\left(pK_\omega - pK_b - \log C\right)$$

In case of substantial hydrolysis the above expression will not be applicable

(iv) Salts of weak base and weak acid

Assuming degree of hydrolysis to be same for the both the ions,

$K_h = K_w / (K_a.K_b)$, $[H^+] = [K_a\, K_w/K_b]^{1/2}$

$$pH = \frac{1}{2}\ (pk_w + pk_a - pk_b)$$

(v) Amphiprotic salts

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e.g. $NaHCO_3$

$$pH = \frac{1}{2}\ (pK_1 + pK_2)$$

pH of such salt solution is independent of concentration od salt.

* ***Buffer solution***

Acidic Buffer

$$pH = pK_a + \log\frac{[salt]}{[acid]}$$

$$\left[\ \text{Actually } pH = pK_a + \log\frac{[\text{conjugate base}]}{[\text{acid}]}\ \right]$$

* Basic buffer

$$pOH = pK_b + \log \frac{[\text{salt}]}{[\text{base}]}$$

$$\left[\text{Actually } pOH = pK_b + \log \frac{[\text{conjugate acid}]}{[\text{base}]} \right]$$

* Simple buffer (Salt of Weak acid & weak base)

$$pH = \frac{1}{2}(pK_w + pK_a - pK_b)$$

* ***Solubility***

For A_xB_y $K_{sp} = (xS)^x\,(yS)^y$

Effect of hydrolysis on solubility $S = \sqrt{K_{sp}\left[1 + \frac{[H^+]}{K_a}\right]}$

* **Indicators**

Indicators	**pH range**		**Colour**
		acid medium	basic medium
Methyl Orange	3.1-4.4	pink	yellow
Phenolphathlene	8.3-10	colourless	pink

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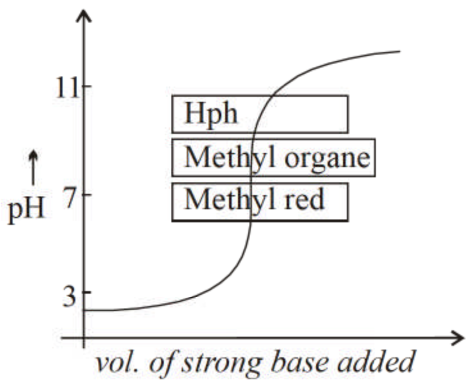
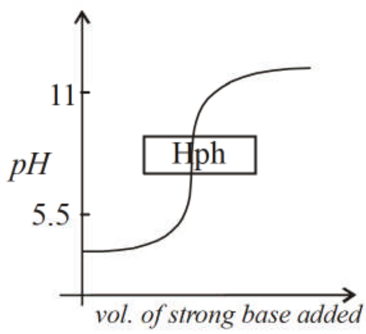
Case I : **Titration of SA + SB**

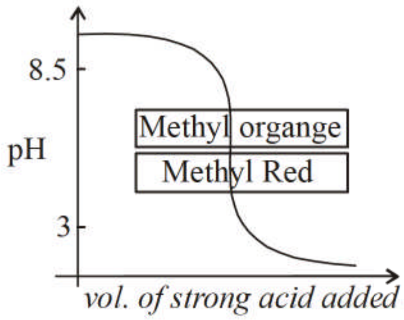


Case II : **Titration of WA + SB**



Methyl orange can't be used. Nothing definite can be said about the use of methyl red.

Case-III : **Titration of WB + SA**



Case-IV : **Titration of WB + WA :** No sharp change in pH. No suitable indicator.

* For complex formation

$$Ag^+ + NH_3 \rightleftharpoons Ag(NH_3)^+ \qquad K_1$$

$$Ag(NH_3)^+ + NH_3 \rightleftharpoons Ag(NH_3)_2^+ \qquad K_2$$

$$Ag^+ + 2NH_3 \rightleftharpoons Ag(NH_3)_2^+ \qquad K_{form} = K_1K_2$$

$$K_d \text{ (dissociation or instability constant)} = \frac{1}{K_{form} \text{ (formation or stability constant)}}$$

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ANSWER KEY

Q.1	D	Q.2	D	Q.3	D	Q.4	B	Q.5	A
Q.6	D	Q.7	B	Q.8	A	Q.9	A	Q.10	D
Q.11	D	Q.12	A	Q.13	C	Q.14	B	Q.15	D
Q.16	A	Q.17	A	Q.18	D	Q.19	C	Q.20	D
Q.21	D	Q.22	A	Q.23	C	Q.24	A	Q.25	B
Q.26	ABCD	Q.27	(A) P (B) R (C) QS			Q.28	0.05	Q.29	0505
Q.30	4860								